Roll No. ....

TEC–502

B. TECH. (ECE) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022

COMMUNICATION SYSTEMS–II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) (i) The spectral range of bandpass signal extends from 10 MHz to 10.4 MHz. Find the minimum sampling rate. (CO1)

(ii) Determine the Nyquist sampling interval, for the given signal : (CO1)

$$x(t) = \sin c\,(700\,t) + \sin c\,(500\,t)$$

OR

(b) State and prove the sampling theorem for lowpass signal. (CO1)

2. (a) Derive an expression for signal to quantization noise ratio for a PCM system which employs uniform quantization technique. Given that input signal to the PCM system is a sinusoidal signal. (CO1)

OR

(b) (i) A 1 kHz sinusoidal signal is ideally sampled at 1500 sample per sec and sampled signal is passed through ideal LPF with a cut-off frequency of 800 Hz. What frequency components will appear in the LPF output ? (CO1)

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(ii) A sinusoidal signal with peak-to-peak amplitude of 1.536 V is quantized into 128 levels using a midrise uniform quantizer. Determine the value of quantization noise power. (CO1)

3. (a) What is companding ? Why is it used ? Illustrate your answer with a sketch of companding curves. (CO1)

OR

(b) Illustrate and describe the types of uniform quantizer. (CO1)

4. (a) Explain Differential Pulse Code Modulation (DPCM) in detail with a suitable diagram. (CO1)

OR

(b) (i) In a binary PCM system, the output signal to quantization noise ratio is to be held to a minimum value of 40 DB. Determine the number of required levels and find the corresponding output signal to quantization noise ratio. (CO1)

(ii) Determine the minimum step-size required for a Delta-Modulation operating at 32 K samples/sec to track the signal (here $u(t)$ is the unit-step function) : (CO1)

$$x(t) = 125\,t\,(u(t) - (t-1)) + (250 - 125\,t)\,(u(t-1) - u(t-2)),$$

so that slope overload is avoided.

5. (a) Draw the block diagram of a digital communication system and explain the function of each block. (CO2)

OR

(b) State and explain the properties of line codes. (CO2)

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